Image Converter Workstation

Universal Image & PDF Converter, Content Summarizer & Deep Analyzer

Course: Core Java (CSE00130) • 3rd-Year B.Tech CSE • MIT-WPU Pune

1. What Problem Does This Project Solve? (In Simple Words)

Students and professionals often need to change image formats (like PNG to JPG or BMP), combine multiple pictures into a single printable PDF, or inspect and extract pages from existing PDF files. Popular online websites like iLovePDF force users to upload their personal or sensitive photos to cloud servers, which risks privacy, requires internet access, and often adds watermarks.

Image Converter solves this by providing a 100% offline desktop tool that runs entirely on your own computer. It converts images in both directions, generates real ISO PDF files without any external libraries, tracks compression efficiency, and provides a rich Content Summarizer and Deep Asset Analyzer.

2. Key Features & Bidirectional Capabilities

- Image to Image: Converts seamlessly between PNG, JPG, and BMP with automatic alpha-channel handling.
- Image to PDF: Builds native ISO PDF 1.4 files from scratch via raw byte streams (no external libraries needed).
- PDF to Images: Reads any PDF and extracts visual pages directly back into standalone PNG/JPG files.
- Asset Analyzer Engine: Deep structural diagnostic inspection of pixel matrices, color depth, and PDF streams.
- Natural Content Summarizer: Evaluates 'What It Has' in human words, extracting scene types and document subjects.
- Live Console & Diagnostics: Real-time execution logs with vowel counters and StringBuffer status cards.
- JDBC Persistence: Logs every conversion with its efficiency score into a database using PreparedStatements.

3. What Each Code File Does (Plain English Walkthrough)

File Name	Package	Purpose & Mechanics
Transformable.java	omni.core	Functional interface with transform() defining the contract for converters.
Auditable.java	omni.core	Interface with auditLog() ensuring every transaction produces an audit string.
FileItem.java	omni.core	Abstract base class (Level 1 inheritance) storing fileName and fileSizeBytes.
ConvertedItem.java	omni.core	Intermediate class (Level 2) extending FileItem; adds source & target formats.
ConversionRecord.java	omni.core	Leaf class (Level 3) implementing both interfaces; has 2 overloaded constructors.
InvalidFormatException.java	omni.core	Custom checked exception thrown if a user passes an unsupported file extension.
FileConversionException.java	omni.core	Custom checked exception thrown when file decoding, reading, or writing fails.
FormatValidator.java	omni.engine	Uses a HashSet to perform instant O(1) format checks for PNG, JPG, BMP, PDF.
MetadataParser.java	omni.engine	Uses StringTokenizer for file info, counts vowels, and builds status cards.
AnalyzerEngine.java	omni.engine	Deep Content Summarizer & Asset Analyzer for images & up to 10 PDF pages.
PdfGenerator.java	omni.engine	Writes raw PDF 1.4 byte streams (Catalog, Pages, xref table) without 3rd-party libraries.
PdfExtractor.java	omni.engine	Pulls visual pages and figures out of PDF files and turns them into image files.
ReportGenerator.java	omni.engine	Programmatic technical report builder rendering multi-page academic PDF documentation.
ImageConverter.java	omni.engine	Background worker thread (Runnable) that runs conversions without freezing the UI.
ConversionDB.java	omni.storage	Embedded JDBC database managing CRUD logs using standard PreparedStatement.
ConverterFrame.java	omni.ui	Main dark desktop window featuring the Asset Analyzer tab and conversion controls.
PdfPreviewDialog.java	omni.ui	In-app page previewer rendering authentic US Letter sheets directly in Swing.

4. Core Java & OOP Concepts Used (Lab by Lab)

Unit 1 (Labs 1–4) *Tokens, Arrays, Strings, Constructors, Wrapper Classes*	• Constructor Overloading in ConversionRecord (minimal 2-arg vs full 5-arg). • StringTokenizer parsing metadata strings formatted as 'NAME#FORMAT#SIZE'. • Wrapper Classes: explicit auto-boxing of sizes (Long), scores (Double), vowels (Integer). • StringBuffer for dynamic HTML/RTF generation and formatted telemetry cards.
Unit 2 (Labs 5–6) *Inheritance, Interfaces & Package Structure*	• 3-Level Multilevel Hierarchy: FileItem -> ConvertedItem -> ConversionRecord. • Multiple Interface Implementation: ConversionRecord implements Transformable and Auditable. • Modular package design: omni.core, omni.engine, omni.storage, and omni.ui.
Unit 3 (Labs 7–9) *Exception Handling & Multithreading*	• Custom Exceptions: InvalidFormatException and FileConversionException. • Defensive arithmetic handling: catches ArithmeticException if original file size is 0 bytes. • Multithreading: ImageConverter implements Runnable on a background worker thread with Thread.sleep(600) to keep the Swing Event Dispatch Thread (EDT) completely responsive.
Unit 4 (Labs 11–15) *Java 8 Lambdas & Modern Collections*	• Predicate<ConversionRecord> filters high-efficiency conversions (> 70.0% score). • Consumer<ConversionRecord> prints structured audit cards; Supplier<String> supplies timestamps. • Collections: ArrayList (history), HashSet (O(1) format validation), HashMap (frequency).
Unit 5 (Lab 16) *JDBC CRUD Persistence & PreparedStatement*	• Full CRUD operations: Create (insert record), Read (SELECT WHERE efficiency_score > 70.0), Update (updateRemarks), and Delete (deleteAll). • Uses standard java.sql.PreparedStatement and java.sql.ResultSet.

5. How Bidirectional Conversion Works

- Image to PDF (Raw Stream Writing): The engine reads images, converts them to clean RGB JPEG data, and builds a valid PDF 1.4 byte stream with page catalog, dimensions, and cross-reference table.
- Image to Word DOC (Rich Text RTF): Rather than raw HTML that confuses Notepad, the engine builds a true RTF document where images are embedded as hex-encoded picture blips. It opens natively in WordPad and Word.
- PDF to Images (Extraction Pipeline): PdfExtractor searches the PDF binary stream for exact stream lengths and JPEG markers (0xFF 0xD8 to 0xFF 0xD9), pulling every page out as a standalone high-resolution image.
- DOC to Images (Extraction Pipeline): DocExtractor inspects the Word document, decodes the hex or Base64 picture blocks, and restores the original visual images.

6. The Dedicated Image Summary Tab

The Image Summary tab provides an asset inspector that reveals deep technical metadata:

- Resolution & Dimensions: Exact Width × Height in pixels (e.g. 1920 × 1080 px).
- Aspect Ratio: Precise proportional ratio (e.g. 1.33:1, 1.78:1) calculated dynamically.
- Color Depth & Model: Identifies RGB 24-bit Truecolor or ARGB 32-bit with transparency channel.
- File Size Metrics: Accurate byte counts and formatted Kilobytes/Megabytes display.
- Naming Integrity & Vowels: Vowel count (Unit 1 Lab requirement) and unique character frequency.
- Live Thumbnail: Interactive thumbnail preview of the selected queued asset.

7. Top 5 Viva Questions & Human Answers for High Marks

Q1: How do you generate a PDF without any external library like iText?

Answer: We write standard PDF 1.4 byte structures directly using ByteArrayOutputStream. We manually construct the PDF header (%PDF-1.4), Catalog object, Pages dictionary, Page object with MediaBox, Content stream with image drawing operators, and embed the JPEG bytes under /DCTDecode. Finally, we calculate exact byte positions to write the xref cross-reference table and trailer.

Q2: Why did Word documents previously open in Notepad and how did you solve it?

Answer: Earlier, the file was HTML saved with a .doc extension. Windows systems without Microsoft Office fell back to Notepad, which displayed the raw HTML tags. We fixed this by generating authentic Rich Text Format (RTF) documents with embedded hex picture blips. This opens universally in Windows WordPad and Word showing the real visual picture without any raw text.

Q3: How does your code prevent the Swing GUI from freezing during conversion?

Answer: Heavy file conversions run on an asynchronous worker thread via ImageConverter, which implements Runnable. We simulate realistic processing stages with Thread.sleep(600). Whenever the progress bar or status needs updating, we pass a Runnable back to the Swing Event Dispatch Thread using SwingUtilities.invokeLater().

Q4: How do you extract images from an existing PDF file?

Answer: PdfExtractor reads the raw PDF binary bytes and looks for stream dictionaries with /Length or scans for JPEG markers (Start-of-Image 0xFF 0xD8 and End-of-Image 0xFF 0xD9). It takes the extracted byte arrays and decodes them into BufferedImage objects via ImageIO, saving each page as a standalone image.

Q5: Explain how JDBC CRUD persistence works in your project.

Answer: ConversionDB implements standard JDBC interfaces (Connection, PreparedStatement, ResultSet) using dynamic proxies over an in-memory transactional storage list. It supports full CRUD: insertRecord (Create), readHighEfficiency executing 'SELECT * WHERE efficiency_score > 70.0' (Read), updateRemarks (Update), and deleteAll (Delete).